Fruit Ripeness Classification Using CNNs

Field

Agriculture and Plant Disease Detection

Project Title

Fruit Ripeness Classification Using CNNs

Background and Motivation

Fruit quality assessment plays a crucial role in agriculture, food processing, and supply chain management. Traditionally, fruit ripeness and freshness are evaluated manually through visual inspection, which is time-consuming, subjective, and prone to human error. Convolutional Neural Networks (CNNs) can automate this process with high accuracy.

Problem Statement

Develop a CNN-based image classification system capable of automatically classifying fruit images into fresh and rotten categories.

Selected Dataset

Fruits Fresh and Rotten for Classification dataset from Kaggle.

Kaggle Notebook Link

https://www.kaggle.com/code/gatlaanshukumar/ml-cnn-model

Dataset Link

https://www.kaggle.com/datasets/sriramr/fruits-fresh-and-rotten-for-classification

Dataset Description

The dataset contains images of fresh and rotten fruits used for image classification.

Number of Images

Training Images: 10,901
Testing Images: 2,695
Total Images: 13,596

Number of Classes

6 Classes:
• Fresh Apples
• Fresh Bananas
• Fresh Oranges
• Rotten Apples
• Rotten Bananas
• Rotten Oranges

Image Type

RGB Color JPEG Images

Class Distribution

Fresh Apples: ~1693
Fresh Bananas: ~1581
Fresh Oranges: ~1466
Rotten Apples: ~2342
Rotten Bananas: ~2224
Rotten Oranges: ~2290

Research Questions

1. Can CNNs accurately classify fruit freshness?
2. Does transfer learning improve classification performance?
3. Which model performs better?

Expected Results

High classification accuracy with MobileNetV2 expected to outperform the custom CNN model.

Proposed Methodology

Dataset Collection → Preprocessing → Resizing & Normalization → Data Augmentation → CNN Training → Evaluation → MobileNetV2 Transfer Learning → Performance Comparison.

Methodology Workflow

Dataset → Preprocessing → Data Augmentation → CNN Model → Training → Evaluation → MobileNetV2 → Comparison

CNN Model Design

Conv2D(32) → MaxPooling → Conv2D(64) → MaxPooling → Conv2D(128) → MaxPooling → Flatten → Dense(256) → Dropout → Softmax Output Layer.

Transfer Learning Model

MobileNetV2 pre-trained on ImageNet with a custom classification head.

Evaluation Metrics

Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and Classification Report.

Results

Custom CNN Accuracy: 97.59%
MobileNetV2 Accuracy: 98.93%

Expected Figures and Tables

Accuracy Curve, Loss Curve, Confusion Matrix, Classification Report, Sample Images, and Model Comparison Table.

Conclusion

The project demonstrates the effectiveness of CNNs and transfer learning for automated fruit ripeness classification.

References

1. Kaggle Fruits Fresh and Rotten Dataset
2. TensorFlow Documentation
3. Keras Documentation
4. MobileNetV2 Research Paper
5. Deep Learning for Image Classification Literature